

Lab 4 – VibeCoding

Chat History / Project Work Log

Project: Reaction-Time Tester (Pygame)

Repository: 43_reaction-time-tester

Purpose: Record the development conversation used to complete the Lab 4 tasks.

1. Project and Task Overview

User: Asked for step-by-step help completing Lab 4 on the assigned Pygame reaction-time tester repository.

Assistant: Explained the four requested tasks: (1) fix reaction-time measurement and false starts, (2) add a game-over/results screen, (3) add replay and difficulty selection, and (4) add sound effects. The work was kept in separate Git commits as required by the lab.

2. Task 1 – Reaction Time and False Starts

User: Worked through the broken reaction-time measurement and asked for implementation help.

Assistant: Updated the Round logic so the valid reaction time is calculated from the exact moment the screen changes to green. An input while the game is still waiting is treated as a false start, with no reaction time recorded.

Implementation principle: waiting → false start/result; go → reaction time/result. False-start reaction times are excluded from the reaction-time list and therefore from the average.

3. Task 2 – Game Over / Results Screen

User: Asked to continue to the next task after fixing the first task.

Assistant: Provided the game-engine implementation for a results screen showing GAME OVER, the reaction time for each valid round, the average reaction time, and an instruction to continue.

User: Successfully committed and pushed Task 2. The recorded commit message was **Add game over results screen**.

4. Task 3 – Replay and Difficulty Selection

Assistant: Added a difficulty-selection screen after the results screen. The proposed settings were Easy (3 rounds, 500–1500 ms wait), Medium (5 rounds, 1000–3000 ms), and Hard (7 rounds, 1500–4000 ms). Space moves from the results screen to difficulty selection, and the selected difficulty starts a new game.

User: Proceeded to Task 4 after completing the replay/difficulty work.

5. Task 4 – Sound Effects

Assistant: Added three sound files under a sounds directory: go.wav, false_start.wav, and game_over.wav. The game engine loads them with pygame.mixer and plays the corresponding sound when the green state begins, when a false start occurs, and when the game ends.

User: Reported that the GO and game-over sounds worked but the false-start sound did not.

Assistant: Changed the input handling so that a None return from register_input() is directly treated as a false start and triggers the false-start sound. A temporary sound-generation script was also provided to regenerate false_start.wav.

User: Reported terminal output showing the false-start sound function was being called successfully, including repeated lines such as “Playing FALSE START sound...” along with “Sounds loaded successfully.”

Assistant: Concluded that the game logic was correctly triggering the sound and that the remaining issue was likely the audio file itself. The user regenerated/tested the sound and then confirmed that the work was completed.

6. False Start Behaviour Discussion

User: Asked whether it is correct for a false start to move immediately to the next round.

Assistant: Explained that in the current implementation a false start counts as a completed round, so the game advances to the next round. The false start is not included in the reaction-time average. Example: if a five-round game has two false starts, only the three valid reaction times contribute to the average.

7. Final Submission Checklist

- Task 1 – Correct reaction timing
- Task 1 – False starts detected
- Task 2 – Results/game-over screen
- Task 2 – Average reaction time
- Task 3 – Replay
- Task 3 – Difficulty selection
- Task 3 – Different difficulty settings
- Task 4 – GO sound
- Task 4 – False-start sound
- Task 4 – Game-over sound
- Separate Git commits for the tasks
- Code pushed to the personal GitHub repository
- Before-changes video
- After-changes video
- Chat history PDF

Note: This PDF is a compiled project chat-history document based on the conversation content available in this session. It is intended for the Lab 4 chat-history deliverable.